

**6 B** The weight of the water converted to ice is the same as that of the ice.

**7 B** Instantaneous power required to drive the escalator

$$= Fv + P_{\text{friction}}$$

$$= Mg \sin \theta \times v + P_{\text{friction}}$$

$$= \left( 62 \times 20 \times 9.81 \times \frac{30}{60} \times 0.75 \right) + 2.5 \times 10^3$$

$$= 7.1 \times 10^3 \text{ W}$$

$$= 7.1 \text{ kW}$$

**8 C**